

Experiment 1:

Write program demonstrates how to use regular expressions in Python to match and search for patterns in text.

Program:

```
import re

text = input("Enter the text: ")

pattern = input("Enter the pattern: ")

result = re.search(pattern, text)

if result:

    print("Pattern Found:", result.group())

else:

    print("Pattern Not Found")
```

Output:

```
import re

text = input("Enter the text: ")
pattern = input("Enter the pattern: ")

result = re.search(pattern, text)

if result:
    print("Pattern Found:", result.group())
else:
    print("Pattern Not Found")

Enter the text: mobile number is 9390727569
Enter the pattern: \d{10}
Pattern Found: 9390727569
```

2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple

